The Ark AI

Bitcoin Lightning AI Service Platform

Software Documentation for Copyright Registration

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1. Abstract

a) Purpose of the Software

The Ark AI is an original software platform that provides artificial intelligence services accessible via Bitcoin Lightning Network micropayments. The platform enables users to access over 140 AI-powered services including text generation, image creation and editing, document generation with automatic PDF export, code assistance, translation across 80+ languages, data analysis, and voice synthesis in multiple African languages.

The software addresses the problem of AI accessibility in developing economies where monthly subscription models are unaffordable for most users. By implementing a pay-per-task micropayment model, The Ark AI makes professional AI tools accessible to anyone with a Bitcoin Lightning wallet.

b) Technologies Used

The Ark AI is built using the following technologies:

- Python 3 - Core backend programming language
- FastAPI - Asynchronous web framework for the REST API
- JavaScript (ES6+) - Frontend interface with real-time streaming
- HTML5 / CSS3 - Responsive web interface
- Server-Sent Events (SSE) - Real-time response streaming
- AI Model APIs - Text and image generation services
- Bitcoin Lightning Network - Micropayment protocol
- Nginx - Reverse proxy with SSL/TLS encryption
- PDF generation libraries - Automated document creation
- Linux server environment - Cloud-hosted deployment

c) Platforms

The Ark AI runs as a web application accessible via any modern web browser on desktop and mobile devices. The software is deployed on a cloud-based virtual private server and is accessible at https://arknode.ai. The platform is designed as a Progressive Web Application (PWA) with service worker support, enabling mobile-app-like experience on smartphones. It supports all major browsers including Chrome, Firefox, Safari, and Edge.

2. Introduction

The Ark AI is a comprehensive artificial intelligence service platform that democratizes access to AI through Bitcoin Lightning micropayments. Unlike traditional AI platforms that require monthly subscriptions, credit cards, and personal account registration, The Ark AI allows anyone to access professional AI services by simply scanning a QR code with a Bitcoin Lightning wallet.

Key Functionalities

- Text AI Services (100+ types): Summarization, translation, content writing, code generation, research, data analysis, legal document drafting, educational content, email composition, and more
- Image Services (24 types): AI image generation, editing, background removal, colorization, photo restoration, style transfer, upscaling, logo design, avatar creation, and more
- Document Generation: Automatic conversion of AI-generated content into professionally formatted, downloadable PDF documents including contracts, business plans, invoices, and more
- Voice Services: Text-to-speech synthesis in multiple languages including African languages (Swahili, Amharic, Zulu, Somali, Afrikaans)
- Translation Services: 80+ language translation with dedicated translation interface
- Chama (Group) AI Access: Collaborative system where teams, schools, or communities pool funds for shared AI usage
- Lightning Wallet System: Built-in wallet allowing users to maintain balances for seamless repeated access
- Developer API: REST API endpoint allowing third-party applications to integrate with the platform
- Free Trial System: New users receive complimentary AI tasks to experience the platform before paying

How the Software Works

1. A user visits the platform and is presented with the main chat interface showing quick-access buttons for common AI tasks.

2. The user types their request or selects a quick-access button. They can also attach images or files for processing.

3. The system automatically detects the task type from the user's input and routes it to the appropriate AI service.

4. For new users, the response streams immediately at no cost as part of the free trial. For subsequent tasks, a Lightning invoice QR code is displayed.

5. The user scans the QR code with any Bitcoin Lightning wallet to pay. Alternatively, users can use the built-in wallet.

6. Upon payment confirmation, the AI response streams in real-time, providing instant feedback.

7. For document tasks, a downloadable PDF is automatically generated. For image tasks, the generated or edited image is displayed with a download button.

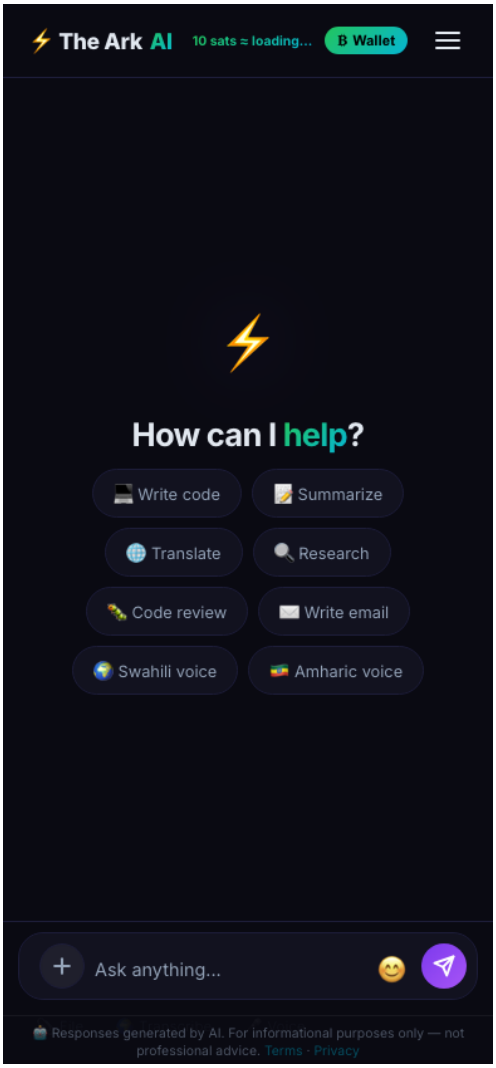
3. Software Documentation

This section provides a sequenced description of the software's key interfaces and functionalities, accompanied by relevant screenshots of the graphical user interface.

3.1 Main Interface - Home Screen

Upon accessing the platform, users are presented with the main chat interface. The home screen features the lightning bolt logo, a greeting message, and quick-access buttons for popular AI services including Write Code, Summarize, Translate, Research, Code Review, Write Email, and African language voice services. The header displays pricing information, a Bitcoin wallet button, and a navigation menu. The chat input field at the bottom allows users to type requests, attach files, or use voice input.

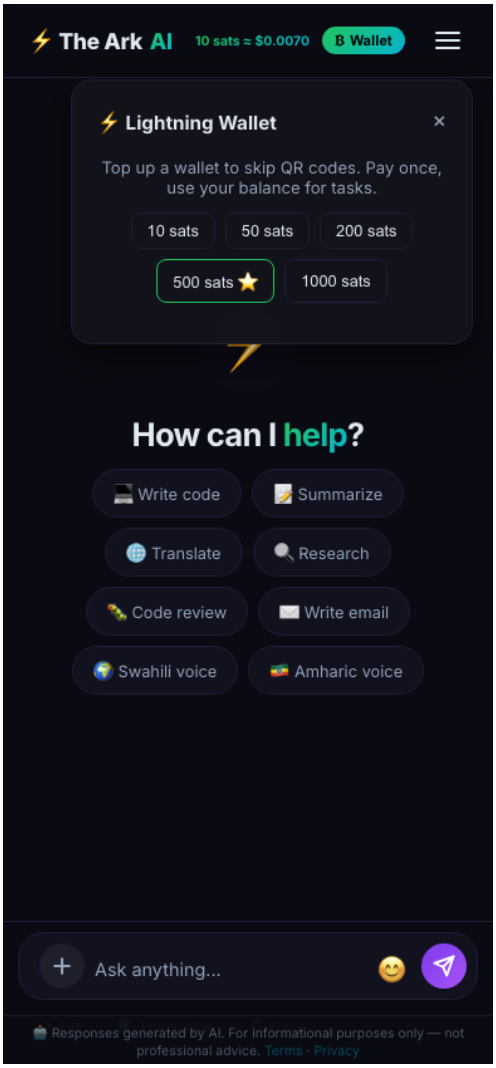
Figure: Main Interface - Home Screen



3.2 Lightning Wallet System

Clicking the Wallet button opens the Lightning Wallet interface. Users can top up their wallet balance by selecting preset amounts. Once topped up, users can access AI services without scanning individual QR codes for each task, providing a seamless prepaid experience. The wallet system tracks balances and deducts costs automatically per task.

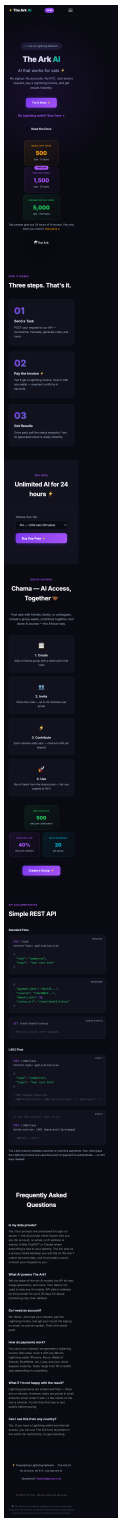
Figure: Lightning Wallet - Top Up Interface



3.3 About Page

The About page provides comprehensive information about the platform, including service tiers, a usage guide, the Chama group AI feature, REST API documentation with code examples, and a Frequently Asked Questions section covering data privacy, payment methods, and service capabilities.

Figure: About Page - Platform Overview



3.4 All Services Marketplace

The All Services page displays all available AI services organized by category. Each service shows its name and brief description. Categories include Code & Development, Text & Strategy, Healthcare & Clinical, Education & Research, Real Estate, Business & Productivity, Agent AI Tools, Voice & Image, and more. Users can browse and select any service to begin a task.

Figure: All Services Marketplace

3.5 Lightning Payment Guide

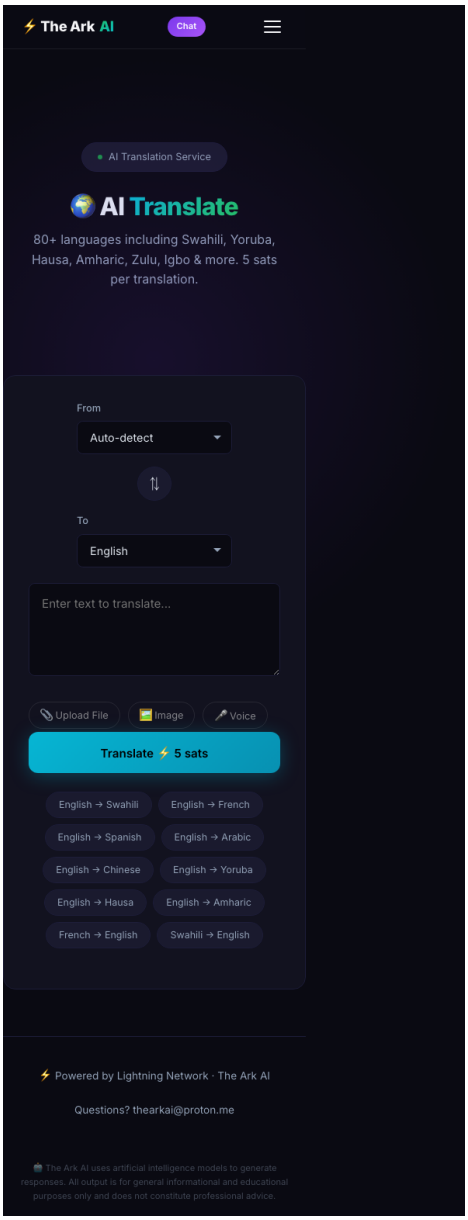
The Lightning Guide page provides a tutorial for users new to Bitcoin Lightning payments. It explains what Bitcoin Lightning is, recommends wallet applications, describes how to acquire Bitcoin, and provides a step-by-step walkthrough of paying for an AI service.

Figure: Lightning Payment Guide

3.6 Translation Interface

The Translate page provides a specialized translation interface supporting 80+ languages with a focus on African languages. The interface features source and target language selectors with auto-detection, a text input area, and options for file, image, or voice input. Quick-access buttons provide one-click translation for popular language pairs.

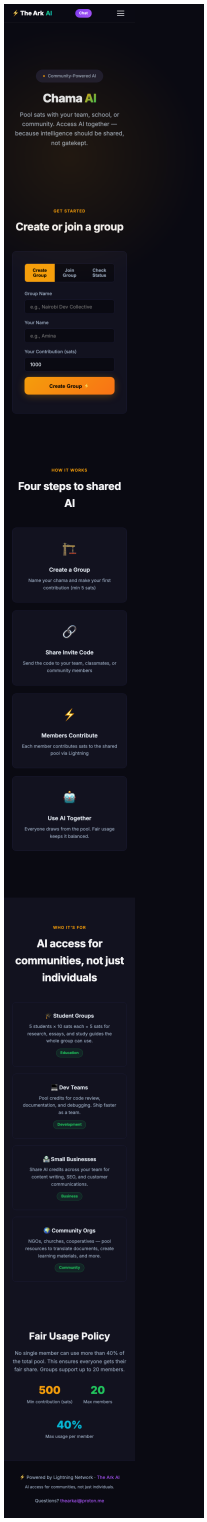
Figure: Translation Interface - 80+ Languages



3.7 Chama - Group AI Access

The Chama feature (Swahili for 'group') enables collaborative AI access for teams, schools, and communities. Users can create a group, share an invite code, and pool contributions for shared AI usage. The system enforces fair usage rules to prevent any single member from consuming a disproportionate share of resources. This feature is designed for student groups, development teams, small businesses, and community organizations.

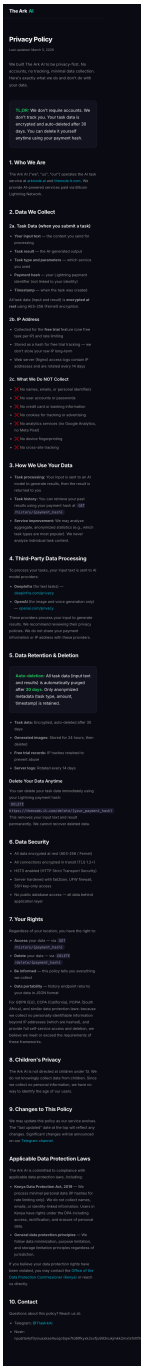
Figure: Chama - Group AI Access



3.8 Privacy Policy

The platform includes a comprehensive privacy policy accessible via the footer. The policy covers data collection practices, usage, storage, and user rights.

Figure: Privacy Policy



The Ark AI

Privacy Policy

Last updated: March 1, 2026

We built The Ark AI to be privacy-first. No accounts, no tracking, minimal data collection. Here's exactly what we do and don't do with your data.

TL;DR: We don't require accounts. We don't track you. Your task data is encrypted and auto-deleted after 30 days. You can delete it yourself anytime using your payment hash.

1. Who We Are

The Ark AI ("we", "us", "our") operates the AI Nish service at arkai.nl and t.me/arkai_bot/app. We provide AI powered services paid via Bitcoin Lightning Network.

2. Data We Collect

2a. Task Data (when you submit a task)

- Your input text — the content you send for processing.
- Task result — the AI generated output
- Task type and parameters — which service you used
- Payment hash — your Lightning payment identifier (not linked to your identity)
- Timestamps — when the task was created

All task data (input and result) is encrypted at rest using AES-256 (Fernet) encryption.

2b. IP Address

- Collected for the free trial fraud prevention (one free task per IP) and rate limiting
- Stored as a SHA-256 hash (not tracking — we don't store your raw IP long-term)
- Web server (Nginx) access logs contain IP addresses and are rotated every 14 days

2c. What We Do NOT Collect

- ❌ No names, emails, or personal identifiers
- ❌ No user accounts or passwords
- ❌ No credit card or banking information
- ❌ No cookies for tracking or advertising
- ❌ No analytics services (no Google Analytics, no Meta Pixel)
- ❌ No device fingerprinting
- ❌ No cross-site tracking

3. How We Use Your Data

- Task processing: Your input is sent to an AI model to generate results. Even the result is returned to you.
- Task history: You can retrieve your past results using your payment hash at GET /history/{payment_hash}
- Service improvement: We may analyze aggregate, anonymized statistics (e.g., which task types are most popular). We never analyze individual task content.

4. Third-Party Data Processing

To provide you tasks, your input text is sent to AI model providers:

- DeepInfra (for text tasks) — deepinfra.com/privacy
- OpenAI (for image and voice generation only) — openai.com/policies/privacy

These providers process your input to generate results. We recommend reviewing their privacy policies. We do not share your payment information or IP address with these providers.

5. Data Retention & Deletion

Auto-deletion: All task data (input text and results) is automatically purged after 30 days. Only anonymized metadata (task type, amount, timestamp) is retained.

- Task data: Encrypted, auto-deleted after 30 days
- Generated images: Stored for 24 hours, then deleted
- Free trial records: IP hashes retained to prevent abuse
- Server logs: Rotated every 14 days

Delete Your Data Anytime

You can delete your task data immediately using your Lightning payment hash:

DELETE https://t.me/arkai_bot/app/history/{your_payment_hash}

This removes your input text and result permanently. We cannot recover deleted data.

6. Data Security

- All data encrypted at rest (AES-256 (Fernet))
- All connections encrypted in transit (TLS 1.2+)
- HSTS enabled (HTTP Strict Transport Security)
- Server hardened with fail2ban, UFW firewall, SSH key-only access
- No public database access — all data behind application layer

7. Your Rights

Regardless of your location, you have the right to:

- Access your data — via GET /history/{payment_hash}
- Delete your data — via DELETE /history/{payment_hash}
- Be informed — this policy tells you everything we collect
- Data portability — history endpoint returns your data in JSON format

For GDPR (EU), CCPA (California), POPIA (South Africa), and similar data protection laws: because we collect no personally identifiable information (beyond IP addresses (which are hashed), and provide full self-service access and deletion, we believe we meet or exceed the requirements of these frameworks.

8. Children's Privacy

The Ark AI is not directed at children under 13. We do not knowingly collect data from children. Since we collect no personal information, we have no way to identify the age of our users.

9. Changes to This Policy

We may update this policy as our service evolves. The "last updated" date at the top will reflect any changes. Significant changes will be announced on our Telegram channel.

Applicable Data Protection Laws

The Ark AI is committed to compliance with applicable data protection laws, including:

- Kenya Data Protection Act, 2019 — We process minimal personal data (IP hashes for rate limiting only). We do not collect names, emails, or identity-linked information. Users in Kenya have rights under the DPA (including access, rectification, and erasure of personal data).
- General data protection principles — We follow data minimization, purpose limitation, and storage limitation principles regardless of jurisdiction.

If you believe your data protection rights have been violated, you may contact the Office of the Data Protection Commissioner (Kenya) or reach us directly.

10. Contact

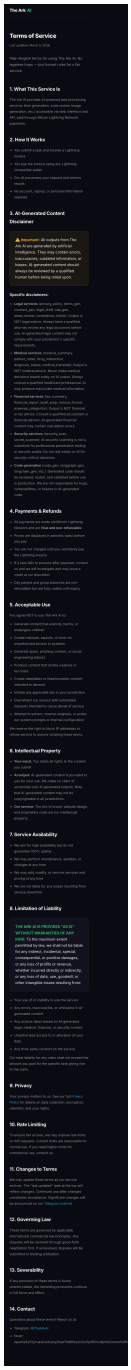
Questions about this policy? Reach us at:

- Telegram: @Tsadmin
- Nostr: npub1ln0f7qmuxnkrkcqz8qe7k3f8fhrzl2c5u5t9l3cxzf4k2cnr5dfm

3.9 Terms of Service

The Terms of Service outlines the legal agreement between users and the platform, covering acceptable use, payment terms, intellectual property, and limitations of liability.

Figure: Terms of Service



The Ark AI

Terms of Service

Last updated: [illegible]

[illegible] terms for using The Ark AI. No [illegible] — just honest rules for a fair service.

1. What This Service Is

The Ark AI provides AI-powered text processing services (text generation, code review, image generation, etc.) accessible via web interface and API, paid through Bitcoin Lightning Network micropayments.

2. How It Works

- You submit a task and receive a Lightning invoice
- You pay the invoice using any Lightning-compatible wallet
- Our AI processes your request and returns results
- No account, signup, or personal information required

3. AI-Generated Content Disclaimer

> ⚠ **Important:** All outputs from The Ark AI are generated by artificial intelligence. They may contain errors, inaccuracies, outdated information, or biases. AI-generated content should always be reviewed by a qualified human before being relied upon.

Specific disclaimers:

- **Legal services** (privacy policy, terms, gdpr, contract, gdpr, legal, draft, nda, gdpr, lease, review, compliance, arbitrator): Output is NOT legal advice. Always have a qualified attorney review any legal document before use. Jurisdiction-specific rules may not apply. [illegible] consultation's specific requirements.
- **Medical services** (medical, symptom, patient, notes, drug, interaction, diagnosis, helper medical, translate): Output is NOT medical advice. Never make medical decisions based solely on AI output. Always consult a qualified healthcare professional. AI may produce inaccurate medical information.
- **Financial services** (tax, invoice, receipt, accounting, bookkeeping, [illegible], format, expense, categorize): Output is NOT financial or tax advice. Consult a qualified accountant or financial advisor. AI-generated financial content may contain calculation errors.
- **Security services** (security, exec, secret, scanner): AI security scanning is not a substitute for professional penetration testing or security audits. Do not rely solely on AI for security-critical decisions.
- **Code generation** (code, gen, snippet, gen, langchain, gen, etc.): Generated code should be reviewed, tested, and validated before use in production. We are not responsible for bugs, vulnerabilities, or failures in AI-generated code.

4. Payments & Refunds

- All payments are made via Bitcoin Lightning Network and are **final and non-refundable**.
- Prices are displayed in satoshis (sats) before you pay.
- You are not charged until you explicitly pay the Lightning invoice.
- If a task fails to execute after payment, contact us and we will investigate and may issue a credit at our discretion.
- Gas pooled and group balances are non-refundable but are fully visible and trackable.

5. Acceptable Use

You agree NOT to use The Ark AI to:

- Generate content that exploits, harms, or endangers children
- Create malware, exploits, or tools for unauthorized access to systems
- Generate spam, phishing content, or social engineering attacks
- Produce content that incites violence or terrorism
- Create deepfakes or impersonation content intended to deceive
- Violate any applicable law in your jurisdiction
- Overwhelm our service with automated requests (intended to cause denial of service)
- Attempt to extract, reverse-engineer, or probe our system prompts or internal configuration

We reserve the right to block IP addresses or refuse service to anyone violating these terms.

6. Intellectual Property

- **Your input:** You retain all rights to the content you submit.
- **Accepted** AI-generated content is provided to you under a CC0 (public domain) or MIT-equivalent license. AI-generated outputs may not be copyrightable in all jurisdictions.
- **Our services:** The Ark AI brand, website design, and proprietary code are our intellectual property.

7. Service Availability

- We aim for high availability but do not guarantee 100% uptime.
- We may perform maintenance, updates, or changes at any time.
- We may add, modify, or remove services and pricing at any time.
- We are not liable for any losses resulting from service downtime.

8. Limitation of Liability

> **THE ARK AI IS PROVIDED "AS IS" WITHOUT WARRANTIES OF ANY KIND.** To the maximum extent permitted by law, we shall not be liable for any indirect, incidental, special, consequential, or punitive damages, or any loss of profits or revenue, whether incurred directly or indirectly, or any loss of data, use, goodwill, or other intangible losses resulting from:

- Your use of or inability to use the service
- Any errors, inaccuracies, or omissions in AI-generated content
- Any actions taken based on AI-generated legal, medical, financial, or security content
- Unauthorized access to or alteration of your data
- Any third-party conduct on the service

Our total liability for any claim shall not exceed the amount you paid for the specific task giving rise to the claim.

9. Privacy

Our privacy policy is in its Ark on SATPrivacy Policy. It covers data collection, encryption, retention, and your rights.

10. Rate Limiting

To ensure fair access, we may impose rate limits on API requests. Current limits are reasonable for normal use. If you need higher limits for commercial use, contact us.

11. Changes to Terms

We may update these terms at any service updates. The "last updated" date at the top will reflect changes. Continued use after changes constitutes acceptance. Significant changes will be announced via our Telegram channel.

12. Governing Law

These terms are governed by applicable international contract law and trader law. Any disputes will be resolved through good-faith negotiation first. If unsuccessful, disputes will be submitted to binding arbitration.

13. Severability

If any provision of these terms is found unenforceable, the remaining provisions continue in full force and effect.

14. Contact

Questions about these terms? Reach us at:

- Telegram: @Yoyoark
- Nostr: [illegible]

4. Key Source Code

The Ark AI software consists of approximately 3,500 lines of original Python source code across 12 backend modules, plus approximately 5,000 lines of HTML/CSS/JavaScript frontend code. The following describes the key source code modules and their functions. Full source code is available upon request by the Kenya Copyright Board for verification purposes.

4.1 Source Code Modules

server.py (2,608 lines) - Core API Server

The main application server module implementing the REST API. Handles HTTP request routing, task type detection and classification, real-time response streaming via Server-Sent Events, Bitcoin Lightning invoice generation and payment verification, user wallet management, free trial tracking, PDF document generation pipeline, image upload and processing, file management, and error handling. This is the central orchestration module that connects all other components.

llm_tasks.py (746 lines) - AI Task Engine

The AI task execution engine that defines and manages over 140 unique service types. Each service has custom prompt engineering tailored to produce optimal results for that specific task category. The module implements intelligent routing between different AI model types (text, image, voice), handles image generation and editing workflows, voice synthesis in multiple languages, and dynamic model selection based on task requirements.

pdf_gen.py (140 lines) - PDF Document Generation

A specialized module that converts AI-generated text content into professionally formatted PDF documents. Implements automatic markdown parsing, branded headers and footers, structured typography with multiple heading levels, bullet point formatting, code block rendering, and automatic file cleanup for expired documents.

wallet.py - Lightning Wallet System

Implements the user wallet system allowing persistent satoshi balance management. Handles wallet creation, balance top-ups via Lightning invoices, per-task balance deductions, and secure storage of wallet data.

openai_compat.py - Developer API Layer

An API compatibility layer that implements the industry-standard /v1/chat/completions endpoint, allowing third-party developer tools and applications to connect to The Ark AI using existing client libraries without modification.

workflows.py - Task Orchestration Engine

A multi-step task orchestration module for complex AI workflows that require sequential processing of multiple operations to produce a final result.

Frontend Code (~5,000 lines)

The web interface code comprising index.html (main chat interface), translate.html (translation interface), groups.html (Chama group access), about.html (platform information), market.html (services marketplace), guide.html (payment tutorial), and supporting pages. Implements responsive design, real-time message streaming, QR code generation for

Lightning payments, image upload with preview, voice input, and Progressive Web App features.

Note: Detailed source code listings are withheld from this public document to protect proprietary implementation details. Complete source code is available for confidential review by the Kenya Copyright Board upon request.

5. Declaration of Originality

I hereby declare that The Ark AI software described in this document is my original work. The source code, architecture, system design, user interface design, prompt engineering, and all creative elements described herein were conceived, designed, and written by me.

The software utilizes open-source libraries and frameworks under their respective permissive licenses (MIT, Apache 2.0), and interfaces with third-party APIs under their commercial terms of service. The original creative work subject to this copyright registration is the unique combination, architecture, implementation, user interface design, and service design built upon these foundations.

This document, the accompanying screenshots of the graphical user interface, and the source code module descriptions constitute the complete submission for copyright registration of The Ark AI software platform under the Kenya Copyright Act, 2001 (Cap 130) as a Literary Work (Computer Program).

Signature: _____

Full Name: _____

ID Number: _____

Date: _____